Syllabus for

Academic year 2020/2021 >

MVE035 - Multivariable analysis

Flervariabelanalys

Syllabus adopted 2019-02-20 by Head of Programme (or corresponding)
Owner: TKTFY
6,0 Credits
Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail
Education cycle: First-cycle
Major subject: Mathematics, Engineering Physics
Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish
Application code: 57115
Open for exchange students: No
Maximum participants: 130
Only students with the course round in the programme plan

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates		
0105	Examination	6,0 c	Grading: TH			6,0 c				Sat 13/03-2021 am J,	Tue 08/06-2021 am J,	Tue 24/08-2021 am J

In programs

TKTFY ENGINEERING PHYSICS, Year 1 (compulsory)

Examiner:

Peter Hegarty

Go to course homepage

Eligibility

General entry requirements for bachelor's level (first cycle)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Linjär algebra och geometri motsvarande kursen TMA660 och *Matematisk analys fortsättning* (envariabelanalys) motsvarande kursen TMA976.

Aim

The course provides basic knowledge of the fundamental theories within mathematical analysis.

Learning outcomes *(after completion of the course the student should be able to)*

The goal is to provide the students with the necessary mathematical tools in multivariable calculus and 3-dimensional vector analysis for subsequent courses in the Engineering Physics and Technical Mathematics programs. AMong the most important learning outcomes are the following:

- To understand the basic concepts of multivariable differential calculus, such as: partial derivative, differentiability, linearization, gradient, implicit and inverse function theorems
- To be able to apply the chain rule to changes of variables in PDE
- To be able to find and classify the stationary points of a multivariable function and apply this knowledge to the solution of optimization problems
- To understand the definition of Riemann integral in arbitrary dimension
- To be able to apply some basic techniques when computing multiple integrals, such as: inspection/symmetry, Fubini's theorem, change of variables, level surfaces

- To be able to handle different parametrizations of curves and surfaces in 3-space, and understand the meaning of and be able to compute line and surface integrals
- To understand Green's theorem in the plane, plus Gauss' and Stokes' theorems in 3-space and apply these to the computation of line and flux integrals
- To acquire some basic knowledge of how the concepts of the course arise in physics, especially in mechanics and electromagnetism
- To be able to differentiate under the integral sign

Content

Functions of several variables. Partial derivatives, differentiability, the chain rule, directional derivative, gradient, level sets, tangent planes.
Taylor's formula for functions of several variables, characterization of stationary points.
Double integrals, iterated integration, change of variables, triple integrals, generalized integrals.
Space curves. Line integrals, Green's formula in the plane, potentials and exact differential forms.
Sufaces in R^3, surface area, surface integrals, divergence and curl, Gauss' and Stokes' theorems.
Some physical problems leading to partial differential equations. Partial differential equations of the first order.
Differentiating under the integral sign.
Functional determinants, inverse functions theorem, implicit functions. Extremal problems for functions of several variables, Lagrange's multiplier rule.

Organisation

The teaching is organized into lectures and exercise sessions (the latter include demonstrations at the blackboard). There are two voluntary items yielding bonus points:

- Computer exercises with Matlab.
- Electronic tests in Möbius.

There are also obligatory blackboard presentations. The students are divided into groups of 6. Each group presents a week of lecture material at the blackboard and writes a report.

Literature

A. Persson, L.-C. Böiers: Analys i flera variabler, Studentlitteratur, Lund.
Övningar till Analys i flera variabler, Institutionen för matematik, Lunds tekniska högskola.

OTHER LITERATURE
L. Råde, B. Westergren: BETA - Mathematics Handbook, Studentlitteratur, Lund.
E. Pärt-Enander, A. Sjöberg: Användarhandledning för Matlab, Uppsala universitet.

Examination including compulsory elements

A written examination.

Bonus point-yielding tests in Möbius.

Bonus point-yielding Matlab problems.

Obligatory blackboard presentations.